COMMERCIAL ITEM DESCRIPTION

# HOLDERS, BORING TOOL, LATHE

The General Services Administration has authorized the use of this commercial item description in preference to the type VI tool holders specified in Federal Specification 00-H-581.

This description covers tool holders that are designed to hold and support boring tools and boring bars used on lathes.

## Salient characteristics.

Classification. This description covers the following types and styles of tool holders, and the tool holder numbers as listed in tables I, II, III, IV, and V.

Type I - Shank mounted.

Style A - Reversible (table I).
Style B - Clamp block (table II).
Style C - Split clamp (table III).

Type II - T-slot mounted.

Style A - Adjustable height (table IV).
Style B - Fixed height (table V).

Dimensional data. The dimensional data of the tool holders shall be as listed in tables I, II, III, IV, and V. The tolerances of all dimensional data shall conform to those specified on the suppliers' parts drawings except where a tolerance range is shown in the tables.

TABLE I. Type I, style A, shank mounted, reversible tool holder.

| Tool holder number | Thickness of shank, inches | Height of shank, inches | Length of shank, inches | Diameter of boring tool accommodated, inches 1/ | | Dimensions of square bit accommodated, inches | |
|---|---|---|---|---|---|---|---|
| | | | | Min. | Max. | Cross Section | Length |
| G-S-10 | 3/8 | 3/4 | 3-3/8 | 1/8 | 1/4 | 1/4 | 2-1/2 |
| G-S-11 | 1/2 | 1 | 3-3/4 | 3/16 | 5/16 | 5/16 | 2-1/2 |
| G-S-12 | 5/8 | 1-1/4 | 4-3/8 | 1/4 | 3/8 | 3/8 | 3 |
| G-S-13 | 3/4 | 1-1/2 | 4-5/8 | 5/16 | 7/16 | 7/16 | 3-1/2 |

1/ See table VI for solid boring tool dimensions.



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TABLE II. Type I, style B, shank mounted, clamp block tool holder.

| Tool holder number | Thickness of shank, inches | Height of shank, inches | Length of shank, inches | Diameter of boring bar accommodated, inches 1/ | |
|---|---|---|---|---|---|
| | | | | Min. | Max. |
| G-M-20 | 3/8 | 3/4 | 3-3/4 | 1/4 | 5/8 |
| G-M-21 | 1/2 | 1-1/8 | 4-5/8 | 1/4 | 3/4 |
| G-M-22 | 5/8 | 1-3/8 | 5-1/2 | 3/8 | 15/16 |
| G-M-23 | 3/4 | 1-5/8 | 6-3/8 | 1/2 | 1-1/8 |

1/ See table VII for dimensions of sleeve type boring bars and square tool bits.

TABLE III. Type I, style C, shank mounted, split clamp holder.

| Tool holder number | Thickness of shank, inches | Height of shank, inches | Length of shank, inches | Diameter of boring bar accommodated, inches 1/ | Applicable adaptable bushing number 2/ |
|---|---|---|---|---|---|
| G-M-29 | 5/16 | 3/4 | 3 | 1/2 | None |
| G-M-30 | 3/8 | 7/8 | 3-1/8 | 9/16 | F-00 |
| G-M-31 | 1/2 | 1-1/8 | 3-1/2 | 3/4 | F-10 & F-11 |
| G-M-32 | 5/8 | 1-3/8 | 4-1/8 | 15/16 | F-20 series |
| G-M-33 | 3/4 | 1-5/8 | 4-3/4 | 1-1/8 | F-30 series |
| G-M-34 | 7/8 | 1-3/4 | 5-1/4 | 1-5/16 | F-40 series |
| G-M-35 | 1 | 2 | 6 | 1-1/2 | F-50 series |

1/ See table VII for dimensions of boring tools specified.
2/ See table VIII for dimensions of adaptable bushings.

TABLE IV. Type II, style A, adjustable height, T-slot tool holder.

| Tool holder number | Diameter of T-slot flange, inches | | Thickness of T-slot flange, inches | | Diameter range of boring bar, inches 1/ | | Span of ring adjustment, inches | |
|---|---|---|---|---|---|---|---|---|
| | Min. | Max. | Min. | Max. | Min. | Max. | Min. | Max. |
| G-L-40 | 2-15/16 | 3-1/16 | 19/32 | 21/32 | 1/2 | 1-5/16 | 1-7/16 | 1-15/16 |
| G-L-41 | 3-7/16 | 3-9/16 | 23/32 | 25/32 | 1/2 | 1-1/2 | 1-11/16 | 2-5/16 |
| G-L-42 | 4-3/16 | 4-5/16 | 27/32 | 29/32 | 5/8 | 1-13/16 | 2-1/16 | 2-7/8 |
| G-L-43 | 4-15/16 | 5-1/16 | 1-1/32 | 1-3/32 | 3/4 | 2-1/4 | 2-3/8 | 3-1/4 |

1/ See table VII for dimensions of boring bars and tool bits.



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TABLE V. Type II, style B, fixed height, T-slot mounted tool holder.

| Tool holder number | Diameter of T-slot flange, inches | | Thickness of T-slot flange, inches | | Width of flange flat, inches | | Diameter of boring bars furnished inches 1/ | | |
|---|---|---|---|---|---|---|---|---|---|
| | Min. | Max. | Min. | Max. | Min. | Max. | Small | Medium | Large |
| G-L-50 | 1-15/16 | 2-1/16 | 11/32 | 13/32 | 1-7/32 | 1-9/32 | 3/8 2/ | 1/2 | 3/4 |
| G-L-51 | 3-1/16 | 3-3/16 | 17/32 | 19/32 | 2-15/32 | 2-17/32 | 1/2 | 3/4 | 1-1/8 |
| G-L-52 | 3-7/16 | 3-9/16 | 21/32 | 23/32 | 2-31/32 | 3-1/32 | 5/8 | 15/16 | 1-5/16 |
| G-L-53 | 4-3/16 | 4-5/16 | 25/32 | 27/32 | 3-15/32 | 3-17/32 | 3/4 | 1-1/8 | 1-1/2 |
| G-L-54 | 4-13/16 | 4-15/16 | 29/32 | 31/32 | 3-31/32 | 4-1/32 | 15/16 | 1-5/16 | 1-13/16 |

1/ See table VII for dimensions of boring bars and tool bits.
2/ A solid boring tool (S-T-6) shall be furnished for the 3/8" diameter boring bar.

Material and hardness. Type I shank mounted tool holders shall be of drop-forged alloy steel, heat-treated to a Rockwell "C" 40 to 52 hardness or alloy steel, surface hardened to a Rockwell "C" 50 to 55 hardness. The type I style C split clamp tool holder head shall be hardened to a Rockwell "C" 26 to 34 hardness. Type II T-slot mounted tool holders shall be of steel.

Description. The tool holders covered herein are described as follows:

Type I, style A. Reversible, shank-mounted boring and turning tool holders shall have a tool holding head grooved on top and bottom for supporting cutting tools with a swiveling yoke style clamping device. The shank shall be off-set 45 degrees from the tool holding head. Two solid boring tools, one square high speed tool steel bit, and a wrench to fit the clamp screw shall be furnished with each tool holder, unless otherwise specified in the contract or purchase order. Dimensions of the tool holder shall conform to table I.

Type I, style B. Clamp block tool holders shall have a reversible clamp block to fasten boring bars to the tool holding head. One sleeve type boring bar of the maximum size accommodated, a wrench to fit the clamp screws and a wrench to fasten tool bits to the boring bar shall be furnished with each holder, unless otherwise specified in the contract or purchase order. Dimensions of the tool holder shall conform to table II.

Type I, style C. Split clamp tool holders shall be furnished with one boring bar of the maximum size within the capacity of the tool holder specified in the contract or purchase order. The



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boring bar may be either a plain bar with two tool bits, a wrench to fit the clamp screw, and a wrench to fit the tool bit fastening device or an end cap bar with three tool bits and a combination wrench to fit the clamp screw and the tool bit clamping device. Adapter bushings for use with the various sizes of boring bars shall be of steel and of the split section type. Dimensions of the tool holders shall conform to table III.

Type II, style A. Each adjustable height tool holder shall be furnished with an adjustable ring for height adjustment of boring tools. One sleeve type bar of the maximum size within the capacity of the holder, two tool bits, wrench to fit the clamp screw, and a wrench to fasten tool bits to the bar shall be furnished with each tool holder unless otherwise specified in the contract or purchase order. Dimensions of the tool holders shall conform to table IV.

Type II, style B. The fixed height tool holder shall accommodate three sizes of boring bars and shall be designed so that a slight turn of the clamp screw will secure or release both bar and holder. Unless otherwise specified in the contract or order, three sleeve bars of appropriante size shall be furnished with each holder. A wrench to fit the clamp screw and a wrench to fasten tool bits to boring bars or a combination wrench to fit both the clamp screw and tool bit fastener device shall be furnished with each tool holder. Dimensions of the tool holders shall conform to table V.

Classification of boring bars and bushings.

S-T series - Solid forged bars with tool numbers and dimensions conforming to table VI.

P-B series - Plain bars having tool bit channels of 90 degrees and 45 degrees at opposite ends of the bar with tool numbers and dimensions conforming to table VII.

S-B series - Sleeve bars having tool bit channels of 90 degrees and 45 degrees at same end of bar 90 degrees apart with tool numbers and dimensions conforming to table VII.

EC-B series - End cap bars having interchangeable end caps with tool bit channels of 45, 60, and 90 degrees with tool number and dimensions conforming to table VII.

F series - Bushings for split clamp holders with tool numbers and dimensions conforming to table VIII.



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TABLE VI. Solid boring tools for type I, style A tool holders.

| Boring tool number | Diameter range of boring tool, inches | Length of boring tool, inches | Width of cutting face, inches | Applicable tool holder number see table I |
|---|---|---|---|---|
| S-T-1 | 1/8 | 4 | 5/16 | G-S-10 |
| S-T-2 | 1/4 | 5 | 17/32 | |
| S-T-3 | 3/16 | 4-1/2 | 7/16 | G-S-11 |
| S-T-4 | 5/16 | 6 | 5/8 | |
| S-T-5 | 1/4 | 5 | 17/32 | G-S-12 |
| S-T-6 | 3/8 | 7 | 3/4 | |
| S-T-7 | 5/16 | 6 | 5/8 | G-S-13 |
| S-T-8 | 7/16 | 8 | 7/8 | |

Measurement system. All dimensions in this description are given in the English system of measurement. These dimensions can be converted to the Metric system equivalents by the use of ANSI/IEEE 268 for Department of Defense use or FED-STD-376 for Civil Agency use.

Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in this description where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Acceptance inspection. The supplier shall randomly select the number of samples specified in table IX that is applicable to the quantity of items being purchased. Each sample shall be visually inspected, measured, and tested, by the supplier, to determine conformance with all requirements specified herein. Any sample failing to meet the salient characteristics specified herein shall be considered a defective item. If the number of defective items equals, or is less than the quantity specified under "Acceptance, number defective" column of table IX, the supplier shall replace the defective samples with acceptable items and the entire lot shall be considered acceptable for shipment. If the number of defective samples equals, or exceeds, the quantity specified in the "Rejection, number defective" column of table IX, the entire lot shall be considered defective and unsuitable for acceptance.



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TABLE VII. Plain, sleeve, and end cap boring bars.

| Number of plain bar | Number of sleeve bar | Number of end cap bar | Diameter of boring bar, inches | Length of boring bar, inches | Square tool bit | | |
|---|---|---|---|---|---|---|---|
| | | | | | Cross section, inches | Length, inches | |
| | | | | | | Straight 90 degree | Angular 45 or 60 degree |
| P-B-1 | S-B-1 | EC-B-1 | 1/2 | 8 | 3/16 | 1 | 2 |
| P-B-2 | - | EC-B-2 | 9/16 | 9 | 3/16 | 1 | 2 |
| P-B-3 | S-B-3 | - | 5/8 | 9-1/8 | 3/16 | 1 | 2 |
| P-B-4 | S-B-4 | EC-B-4 | 3/4 | 11 | 1/4 | 1-7/32 | 2-1/2 |
| P-B-5 | S-B-5 | EC-B-5 | 15/16 | 14 | 5/16 | 1-7/16 | 2-1/2 |
| P-B-6 | S-B-6 | EC-B-6 | 1-1/8 | 16 | 3/8 | 1-13/16 | 3 |
| P-B-7 | S-B-7 | - | 1-5/16 | 18 | 3/8 | 1-13/16 | 3 |
| P-B-8 | - | EC-B-8 | 1-5/16 | 18 | 7/16 | 2-3/16 | 3-1/2 |
| P-B-9 | S-B-9 | - | 1-1/2 | 23 | 7/16 | 2-3/16 | 3-1/2 |
| P-B-10 | - | EC-B-10 | 1-1/2 | 23 | 1/2 | 2-1/2 | 4 |
| P-B-11 | S-B-11 | - | 1-13/16 | 28 | 1/2 | 2-1/2 | 4 |
| P-B-12 | - | EC-B-12 | 1-13/16 | 28 | 5/8 | 3-1/4 | 4-1/2 |
| P-B-13 | S-B-13 | - | 2-1/4 | 34 | 5/8 | 3-1/4 | 4-1/2 |
| P-B-14 | - | EC-B-14 | 2-1/4 | 31 | 3/4 | 4-3/4 | 5 |



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TABLE VIII. Bushings for split clamp holders.

| Adapter bushing number | Outside diameter, inches | Inside diameter, inches | Length of bushings, inches | |
|---|---|---|---|---|
| | | | Min. | Max. |
| F-00 | 9/16 | 1/2 | 1-11/16 | 1-13/16 |
| F-10 | 3/4 | 1/2 | 2-1/16 | 2-3/16 |
| F-11 | 3/4 | 9/16 | 2-1/16 | 2-3/16 |
| F-20 | 15/16 | 1/2 | 2-7/16 | 2-9/16 |
| F-21 | 15/16 | 9/16 | 2-7/16 | 2-9/16 |
| F-22 | 15/16 | 3/4 | 2-7/16 | 2-9/16 |
| F-30 | 1-1/8 | 1/2 | 2-13/16 | 2-15/16 |
| F-31 | 1-1/8 | 9/16 | 2-13/16 | 2-15/16 |
| F-32 | 1-1/8 | 3/4 | 2-13/16 | 2-15/16 |
| F-33 | 1-1/8 | 15/16 | 2-13/16 | 2-15/16 |
| F-40 | 1-5/16 | 3/4 | 3-3/8 | 3-1/2 |
| F-41 | 1-5/16 | 15/16 | 3-3/8 | 3-1/2 |
| F-42 | 1-5/16 | 1-1/8 | 3-3/8 | 3-1/2 |
| F-50 | 1-1/2 | 15/16 | 3-13/16 | 3-15/16 |
| F-51 | 1-1/2 | 1-1/8 | 3-13/16 | 3-15/16 |
| F-52 | 1-1/2 | 1-5/16 | 3-13/16 | 3-15/16 |

TABLE IX. Sampling plan.

| Quantity of items being purchased | Sample size | Acceptance, number defective | Rejection, number defective |
|---|---|---|---|
| 2-8 | 2 | 0 | 1 |
| 9-15 | 3 | 0 | 1 |
| 16-25 | 5 | 0 | 1 |
| 26-50 | 8 | 1 | 2 |
| 51-90 | 13 | 1 | 2 |
| 91-150 | 20 | 2 | 3 |
| 151-280 | 32 | 3 | 4 |
| 281-500 | 50 | 5 | 6 |

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.



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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified in the contract or order, preservation, packaging, packing, labeling, and marking shall be in accordance with ASTM D 3951. When specified in the contract or order, preservation, packaging, packing, labeling, and marking shall be in accordance with Federal Specification PPP-T-1150, level as specified.

Notes. Purchaser should specify the following:

a. Title, number, and date of this commercial item description.
b. Quantity, type, style, and tool holder number.
c. Boring bars, cutter bits, bushings, and wrenches required if not normally furnished.
d. Specify packaging in accordance with PPP-T-1150 and level, if required.

Boring tools, cutter bits, bushings, and wrenches, normally furnished by the supplier will be furnished, unless otherwise specified in the contract or order.

Type, style, and tool numbers of this description correspond to the class, style, and tool numbers of type VI tool holders in Federal Specification 00-H-581.

ANSI/IEEE 268 "Metric Practice" is available from American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.

ASTM D 3951 "Packaging, Commercial" is available from American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.

FED-STD-376 "Preferred Metric Units for General Use by the Federal Government" and Federal Specification PPP-T-1150 "Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery, Packaging and Packing of" should be obtained from the procuring activity or as directed by the contracting officer.

MILITARY INTERESTS:

Custodians

Army - AL
Navy - SH
Air Force - 99

Review Activities

Air Force - 84
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

DLA - IP

Project 3460-0115



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